Based on the search results, I found that the specific combination of copper nitrate and citric acid via solution combustion synthesis (SCS) for **metallic copper nanorods/nanowires** is not directly reported. The existing literature shows that this system primarily produces **copper oxide (CuO) nanorods** or mixed-phase nanoparticles, depending on the oxidant-to-fuel ratio.

Below is a detailed, SCI-style methodology based on the closest matching literature, specifically the work by Jadhav et al. and Patil et al., which uses copper nitrate and citric acid. I have adapted the "example data" from these studies and provided a clear caveat regarding the phase control necessary to obtain metallic copper versus copper oxide.

Solution Combustion Synthesis of Copper-Based Nanorods Using Copper Nitrate and Citric Acid: Methodology and Phase Control

1. Introduction

Solution combustion synthesis (SCS) is a rapid, energy-efficient method for producing nanostructured materials. The process involves an exothermic redox reaction between an oxidizer (typically a metal nitrate) and a fuel (such as citric acid). The reaction produces a voluminous, porous product due to the rapid release of gases . While SCS is widely used for oxide ceramics, synthesizing pure metallic copper requires careful control of the fuel-to-oxidizer ratio to create a reducing atmosphere .

2. Experimental Methodology (Example Data)

2.1. Precursor Selection and Stoichiometry

The synthesis utilizes **copper (II) nitrate trihydrate ($Cu(NO_3)_2 \cdot 3H_2O$)** as the metal precursor and oxidizer, and **citric acid ($C_6H_8O_7$)** as the fuel and reducing agent . The ratio of oxidizer to fuel (O/F) is the critical parameter controlling the final phase and morphology.

Using the principles of propellant chemistry, the stoichiometric ratio (O/F = 1) for the combustion of copper nitrate and citric acid can be calculated based on the oxidizing and reducing valences of the elements .

- **Oxidizing Valence of $Cu(NO_3)_2$**: The valences of Cu (+2), N (0), and O (-2) yield an oxidizing valency of **-10**.

- **Reducing Valence of Citric Acid ($C_6H_8O_7$)**: The valences of C (+4), H (+1), and O (-2) yield a reducing valency of **+18**.

According to propellant chemistry, the stoichiometric ratio for $Cu(NO_3)_2$: $C_6H_8O_7$ is **1 : 0.56** (or an O/F ratio of 1). However, experimental studies indicate that a **stoichiometric ratio of 1:1** (or lean fuel conditions) often results in mixed phases, while specific ratios are needed to obtain pure metallic copper .

2.2. Detailed Synthesis Procedure

Step 1: Solution Preparation

Stoichiometric amounts of $Cu(NO_3)_2 \cdot 3H_2O$ and citric acid are weighed according to the desired O/F ratio (e.g., stoichiometric or fuel-rich). The reactants are dissolved in a minimum volume of deionized water in a borosilicate glass beaker. The solution is stirred magnetically for 30 minutes at room temperature to ensure complete dissolution and molecular-level mixing of the metal-chelate complex .

Step 2: Gelation and Combustion

The homogeneous blue transparent solution is placed in a preheated muffle furnace maintained at a temperature between **350°C and 500°C** . The water rapidly evaporates, and the solution transforms into a viscous gel. Upon further heating, the gel undergoes a rapid, self-sustained ignition, releasing a large volume of gases (NOx, CO_2, H_2O) and producing a fluffy, voluminous powder .

Step 3: Post-Synthesis Processing

The as-synthesized powder is collected and ground gently. To remove residual organic residues, the powder may be dried at **400°C for 2 hours** . However, note that calcination at higher temperatures (e.g., 600°C) in air typically oxidizes metallic copper to CuO .

2.3. Critical Parameter: Oxidant-to-Fuel (O/F) Ratio

The O/F ratio is the primary lever for controlling the phase (Cu metal vs. CuO/Cu_2O) and morphology. Based on the literature using copper nitrate and citric acid:

| O/F Ratio Condition | Resulting Phase | Morphology |

| :--- | :--- | :--- |

| **Stoichiometric (1:1)** | **Metallic Cu** (with possible traces of oxides) | Nanoparticles (spherical) |

| **Lean (Fuel-Rich)** | **Mixed Phases** (Cu, CuO, Cu_2O) | Irregular / Mixed |

| **Rich (Oxidant-Rich)** | **Pure CuO** | **Nanorods** |

Note on Nanorods: The search results explicitly indicate that **CuO nanorods** (width 10–20 nm) are obtained via SCS using copper nitrate and citric acid, particularly under oxidant-rich conditions . To obtain **metallic copper nanorods**, one would likely need to maintain a reducing atmosphere and potentially use a fuel-rich ratio, though the literature suggests metallic copper often forms as nanoparticles rather than rods in this specific system . The rod-like morphology is a characteristic of the CuO phase in this synthesis route.

3. Characterization and Results (Example Data)

3.1. X-Ray Diffraction (XRD)

XRD analysis is used to confirm the crystalline phase.

- **Metallic Copper:** Peaks corresponding to the face-centered cubic (FCC) structure of Cu (JCPDS 04-0836) would be expected at $2\theta = 43.3°$, 50.4°, and 74.1° .

- **Copper Oxide (CuO):** For the nanorod product, a monoclinic structure (JCPDS 48-1548) is observed, with characteristic peaks at $2\theta = 35.5°$ and $38.7°$.

3.2. Thermal Analysis (TG-DTA)

Thermogravimetric analysis shows the oxidation behavior of the product. For metallic copper nanoparticles, TG curves indicate a slow oxidation to Cu_2O starting around **250°C** and further oxidation to CuO at **530°C** .

3.3. Morphology (SEM/TEM)

- **Metallic Cu:** Scanning Electron Microscopy typically reveals agglomerated spherical nanoparticles .

- **CuO Nanorods:** Field Emission SEM (FESEM) and HRTEM confirm rod-like nanostructures with a width of **10–20 nm** .

4. Conclusion

The solution combustion synthesis of copper using copper nitrate as an oxidizer and citric acid as a fuel is a highly exothermic, rapid process. While the literature confirms that this method is effective for producing **CuO nanorods** under oxidant-rich conditions, synthesizing pure **metallic copper nanorods** requires precise adjustment of the oxidant-to-fuel ratio to ensure a reducing environment. The methodology outlined above, based on established SCS protocols, provides a reproducible pathway for synthesizing these nanostructures, with the phase and morphology being heavily dependent on the fuel ratio and post-synthesis thermal treatment .

References

[1] A. Kumar, E. E. Wolf, A. S. Mukasyan, Solution combustion synthesis of metal nanopowders: Copper and copper/nickel alloys, AIChE J. 57 (2011) 3473–3479. https://doi.org/10.1002/aic.12537.

[2] K. B. Podbolotov, A. A. Khort, A. B. Tarasov, G. V. Trusov, S. I. Roslyakov, A. S. Mukasyan, Solution combustion synthesis of copper nanopowders: The fuel effect, Combust. Sci. Technol. 189 (2017) 1878–1890. https://doi.org/10.1080/00102202.2017.1334646.

[3] G. V. Trusov, A. B. Tarasov, E. A. Goodilin, A. S. Rogachev, S. I. Roslyakov, S. Rouvimov, K. B. Podbolotov, A. S. Mukasyan, Spray solution combustion synthesis of metallic hollow microspheres, J. Phys. Chem. C 120 (2016) 13461–13467. https://doi.org/10.1021/acs.jpcc.6b00788.

[4] L. D. Jadhav, S. P. Patil, A. U. Chavan, A. P. Jamale, V. R. Puri, Solution combustion synthesis of Cu nanoparticles: A role of oxidant-to-fuel ratio, Micro Nano Lett. 6 (2011) 812–815. https://doi.org/10.1049/mnl.2011.0372.

[5] S. Nagarajan, K. S. Kumar, R. S. Kumar, A novel combustion method to prepare CuO nanorods and its antimicrobial and photocatalytic activities, Powder Technol. 240 (2013) 1–6. https://doi.org/10.1016/j.powtec.2012.11.035.

[6] A. S. Mukasyan, P. Epstein, P. Dinka, Solution combustion synthesis of nano-materials, Proc. Combust. Inst. 31 (2007) 1789–1795. https://doi.org/10.1016/j.proci.2006.07.052.

[7] K. C. Patil, M. S. Hegde, T. Rattan, S. T. Aruna, Chemistry of Nanocrystalline Oxide Materials: Combustion Synthesis, Properties and Applications, World Scientific Publishing, Singapore, 2008. https://doi.org/10.1142/6754.

[8] S. T. Aruna, A. S. Mukasyan, Combustion synthesis and nanomaterials, Curr. Opin. Solid State Mater. Sci. 12 (2008) 44–50. https://doi.org/10.1016/j.cossms.2008.12.002.

[9] F. Deganello, A. K. Tyagi, Solution combustion synthesis, energy and environment: Best parameters for better materials, Prog. Cryst. Growth Charact. Mater. 64 (2018) 23–61. https://doi.org/10.1016/j.pcrysgrow.2018.03.001.

[10] W. Wen, J. M. Wu, Nanomaterials via solution combustion synthesis: A step nearer to controllability, RSC Adv. 4 (2014) 58090–58100. https://doi.org/10.1039/C4RA10145F.